
Medical Image Classification for Early Melanoma Detection

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Abstract

Melanoma is one of the deadliest skin cancers, but early detection dramatically improves patient outcomes. We propose a supervised learning system that classifies dermoscopic images of skin lesions as benign or malignant, prioritizing sensitivity to reduce false negatives while maintaining acceptable specificity to limit unnecessary biopsies. We benchmark multiple transfer-learned CNN backbones (ResNet-50, EfficientNet, DenseNet, MobileNetV2) and systematically compare class-imbalance handling strategies, including weighted loss, focal loss, oversampling, and data augmentation. Models are evaluated using AUC-ROC, balanced accuracy, and clinically meaningful operating points such as sensitivity at fixed specificity. Our final deliverables include a comprehensive evaluation report and a lightweight web demo for image upload and model inference.

1 Problem Definition

1.1 Lay-Person Version

Skin cancer is the most common cancer globally, and melanoma is among its most dangerous forms. Early detection substantially improves survival: the five-year survival rate for localized melanoma exceeds 98%, while metastatic cases drop to roughly 30%. Clinical diagnosis is difficult even for trained dermatologists, and access to specialists is uneven. We aim to build a model that screens dermoscopic lesion images and flags likely malignancies, serving as a decision-support tool to reduce missed diagnoses.

1.2 Formal Definition

We study a binary image classification problem. Given an input image

$$x \in \mathbb{R}^{224 \times 224 \times 3},$$

we wish to learn a function

$$f_{\theta}(x) \rightarrow \{0, 1\}$$

where 0 denotes benign and 1 denotes malignant. This is a supervised learning problem with two complicating factors: (1) significant class imbalance, since benign lesions are far more prevalent in practice, and (2) asymmetric error costs, since false negatives (missed cancers) carry a substantially higher clinical cost than false positives.

Our primary training objective is a weighted cross-entropy loss that penalizes false negatives more heavily:

$$\mathcal{L}_{\text{WCE}} = -w_1 y \log p - w_0 (1 - y) \log(1 - p),$$

where $p = f_\theta(x)$ is the predicted probability of malignancy, $y \in \{0, 1\}$ is the ground-truth label, and $w_1 > w_0$ reflects the higher cost of missing a true positive. We additionally explore focal loss as an alternative objective (Section 3).

2 Related Work

Prior work demonstrates that deep learning can match or exceed dermatologist performance on curated skin lesion datasets, with progress driven by transfer learning, class-imbalance handling, architectural innovation, and large-scale benchmark challenges. Below we review the eight most relevant contributions, identify their strengths and limitations, and explain how our work addresses those limitations.

2.1 Esteva et al. (2017) — Dermatologist-Level Classification

Esteva et al. demonstrated dermatologist-level skin lesion classification using an Inception v3 CNN pretrained on ImageNet and fine-tuned on over 129,000 labeled clinical images (1). The model was evaluated against 21 board-certified dermatologists on two binary classification tasks.

Strengths: This work established transfer learning from large-scale natural image datasets as an effective pipeline for skin lesion recognition, and showed that a single CNN trained end-to-end could generalize across a broad taxonomy of skin diseases.

Weaknesses: Evaluation was conducted on a curated, controlled test set, raising questions about real-world generalizability. The study used a single architecture, predates modern imbalance-mitigation techniques such as focal loss, and does not report sensitivity at clinically fixed specificity thresholds.

How we address it: We adopt the same ImageNet pretraining paradigm but systematically compare multiple modern backbone families rather than committing to a single architecture. We also report sensitivity at fixed specificity to enable clinically grounded comparisons.

2.2 Tschandl et al. (2018) — The HAM10000 Dataset

Tschandl et al. curated the HAM10000 dataset, consisting of 10,015 dermoscopic images spanning 7 diagnostic categories (3). It became the field’s de facto benchmark and served as the basis for the ISIC 2018 Challenge.

Strengths: HAM10000 is large, publicly accessible, and includes rich metadata (patient age, sex, lesion localization, and acquisition method), enabling reproducible research and systematic comparisons across methods.

Weaknesses: The dataset exhibits severe class imbalance: melanocytic nevi account for the vast majority of samples, while categories such as dermatofibroma are severely underrepresented. This imbalance can cause models to be heavily biased toward majority classes. The dataset also has known limitations in skin-tone diversity, restricting generalizability across patient populations.

How we address it: We directly target class imbalance through a controlled comparison of weighted cross-entropy, focal loss, and oversampling. We discuss fairness limitations explicitly and, where metadata permits, report stratified performance across demographic subgroups.

2.3 Haenssle et al. (2018) — Man vs. Machine

Haenssle et al. benchmarked a deep CNN against 58 international dermatologists on dermoscopic melanoma recognition, finding that the CNN matched or exceeded physician-level performance at the same sensitivity operating points (2).

Strengths: By comparing against practicing clinicians and reporting performance at fixed sensitivity and specificity thresholds, this work grounded CNN evaluation in clinically realistic terms, rather than reporting only aggregate AUC scores.

Weaknesses: The study evaluated a single architecture on a single held-out test set. It did not explore imbalance mitigation, augmentation choices, or lightweight deployment for resource-constrained settings.

How we address it: We adopt sensitivity at fixed specificity as a core evaluation metric, extending the analysis across multiple architectures and imbalance strategies, and include MobileNetV2 as a deployment-friendly option.

2.4 He et al. (2016) — Deep Residual Networks

He et al. introduced ResNets, which use skip connections to address the vanishing gradient problem, enabling the training of very deep networks and achieving state-of-the-art results on ImageNet (4).

Strengths: ResNets demonstrated that residual connections allow networks to learn identity mappings, making depth tractable. ResNet-50 remains a strong, well-understood baseline across a wide range of vision tasks, including medical image classification.

Weaknesses: ResNet architectures can be parameter-heavy relative to their performance when compared to more recent compound-scaling models. They were designed for natural images and may need careful fine-tuning for dermoscopic data.

How we address it: We use ResNet-50 as our primary baseline, comparing it directly against more modern architectures to measure the practical gains available from architectural advances in the skin lesion classification setting.

2.5 Huang et al. (2017) — Densely Connected Networks

Huang et al. proposed DenseNet, in which each layer receives direct connections from all preceding layers (5). This dense connectivity encourages feature reuse and substantially reduces the number of parameters relative to ResNets of comparable depth.

Strengths: DenseNet’s feature reuse improves gradient flow and parameter efficiency, and it has shown strong performance on medical imaging tasks where labeled data is limited. It is also naturally resistant to overfitting due to implicit deep supervision.

Weaknesses: Dense connectivity increases memory consumption during training, since all intermediate feature maps must be retained. This can be a practical constraint in GPU-limited settings.

How we address it: We include DenseNet-121 in our benchmark suite and measure its accuracy-efficiency trade-off under our hardware constraints, specifically evaluating whether its reduced parameter count translates to faster training or better generalization on our imbalanced dataset.

2.6 Tan & Le (2019) — EfficientNet

Tan and Le proposed EfficientNet, which scales CNN width, depth, and resolution simultaneously via a principled compound coefficient (6). EfficientNet-B0 through B7 provide a family of models spanning a wide range of accuracy-versus-compute trade-offs.

Strengths: EfficientNets achieve higher accuracy than comparably sized ResNets and DenseNets on ImageNet, and their compound scaling makes it straightforward to select a model appropriate for a given compute budget.

Weaknesses: EfficientNets can be sensitive to the choice of input resolution and regularization scheme, and their performance advantage over simpler architectures is not always maintained when fine-tuning on small domain-specific datasets.

How we address it: We benchmark EfficientNet-B0 through B2 and evaluate whether the ImageNet accuracy advantage transfers to dermoscopic lesion classification under our fixed augmentation and imbalance-mitigation pipeline.

2.7 Lin et al. (2017) — Focal Loss

Lin et al. introduced focal loss as a modification of standard cross-entropy that down-weights well-classified easy examples and focuses learning on hard, misclassified examples (9). The loss is defined

as:

$$\mathcal{L}_{\text{FL}} = -\alpha_t(1 - p_t)^\gamma \log(p_t),$$

where p_t is the model’s estimated probability for the true class, α_t is a class-balancing weight, and $\gamma \geq 0$ is a tunable focusing parameter.

Strengths: Focal loss provides a principled, differentiable mechanism for addressing class imbalance and mining hard examples without requiring explicit resampling, and has shown strong results across domains beyond its original object detection setting.

Weaknesses: The γ focusing parameter requires careful tuning, and the relative benefit over weighted cross-entropy is dataset- and architecture-dependent. It may not outperform simpler strategies in all settings.

How we address it: We treat focal loss as one of four imbalance-mitigation strategies in a controlled ablation study, measuring its benefit specifically on skin lesion data against weighted cross-entropy and oversampling baselines.

2.8 Zhang et al. (2018) and DeVries & Taylor (2017) — Augmentation Strategies

Zhang et al. proposed MixUp (10), a training strategy that forms convex combinations of input image pairs and their labels to improve generalization and calibration. DeVries and Taylor introduced CutOut (11), which randomly masks contiguous rectangular patches to force the model to use distributed rather than localized features.

Strengths: Both techniques are architecture-agnostic, computationally cheap, and have shown consistent generalization improvements across image classification benchmarks. MixUp also improves probability calibration, which is especially valuable in a medical decision-support context.

Weaknesses: Their benefit on dermoscopic images is not well characterized, and naive application of MixUp to skin lesion images can produce visually implausible blends that may not reflect the data distribution. Lesion-specific augmentations, such as hair artifact simulation, may be more domain-appropriate.

How we address it: We include both MixUp and CutOut in a standardized augmentation pipeline applied uniformly across all models, so architectural comparisons are not confounded by augmentation differences. We treat their combined use as one condition in our ablation study.

3 Algorithms, Techniques, and Models

3.1 Model Architectures

All models are initialized from ImageNet-pretrained weights and adapted for binary classification by replacing the final fully connected layer with a single sigmoid output unit. We benchmark:

- **ResNet-50** (4) — strong residual baseline.
- **EfficientNet-B0, B2** (6) — compound-scaled efficiency-oriented models.
- **DenseNet-121** (5) — densely connected, parameter-efficient network.
- **MobileNetV2** (7) — lightweight model for deployment-constrained inference.
- **Vision Transformer (ViT-Base)** (8) — attention-based stretch goal.

3.2 Transfer Learning Strategy

We use a two-phase fine-tuning procedure. In Phase 1, the backbone is frozen and only the classification head is trained for 5 epochs using a learning rate of 10^{-3} . In Phase 2, the entire network is unfrozen and trained end-to-end for up to 20 additional epochs with a reduced learning rate of 10^{-4} and cosine annealing. Early stopping based on validation AUC is applied in both phases.

3.3 Class Imbalance Handling

We conduct a controlled ablation across four strategies:

- **Weighted cross-entropy:** class weights set to the inverse of class frequencies.
- **Focal loss:** $\gamma \in \{0.5, 1, 2\}$ tuned on the validation set.
- **Oversampling:** random oversampling of the minority (malignant) class to achieve a 1:2 ratio.
- **Combined:** focal loss with simultaneous oversampling.

3.4 Augmentation Pipeline

A fixed augmentation pipeline is applied to all models during training: random horizontal and vertical flips, rotations over $[0^\circ, 360^\circ]$, color jitter (brightness, contrast, saturation), CutOut, and MixUp ($\alpha = 0.4$). At test time, only center-cropping and normalization are applied.

4 Evaluation Methods

4.1 Primary Metrics

- **Sensitivity / Recall** — primary clinical objective; minimizes missed malignancies.
- **AUC-ROC** — threshold-independent summary of discriminative performance.
- **Balanced accuracy** — arithmetic mean of sensitivity and specificity; accounts for class imbalance.

4.2 Secondary Metrics and Analyses

- Specificity, precision, and F1-score.
- Per-class confusion matrices and error analysis.
- Sensitivity at fixed specificity of 80% and 90%, to reflect clinically relevant operating points.
- Calibration curves (reliability diagrams) to assess the trustworthiness of predicted probabilities.

4.3 Testing Protocol

The dataset is partitioned into 70% train, 10% validation, and 20% test splits using stratified sampling to preserve class ratios. The held-out test set is not used for any tuning or model selection decisions. Where computationally feasible, we use 5-fold cross-validation on the training set for hyperparameter selection. If time permits, we will apply McNemar’s test for pairwise model comparisons and the DeLong test for AUC comparisons.

5 Data

5.1 Source and Size

We will use the “Skin Cancer: Malignant vs. Benign” dataset sourced from the ISIC Archive and hosted on Kaggle. The dataset contains approximately **13,500 RGB images** in JPEG format, split across benign and malignant classes, with a roughly 4:1 imbalance in favor of benign lesions. All images are resized to 224×224 pixels prior to training.

5.2 Preprocessing and Known Issues

- **Normalization:** pixel values are scaled to $[0, 1]$ and standardized using ImageNet channel means and standard deviations.
- **Imbalance:** benign lesions are heavily overrepresented; addressed via the strategies in Section 3.
- **Duplicates:** the ISIC archive contains known near-duplicates across collection years; we will perform perceptual hash-based deduplication before splitting.

- **Fairness:** dermoscopic datasets are known to underrepresent darker skin tones. We will explicitly discuss this limitation and, where demographic metadata is available, report stratified performance.

6 Expected Accomplishments

By the end of the semester, we expect to deliver:

- A controlled benchmark comparing 4–5 architectures under identical training and evaluation conditions.
- A systematic ablation study of imbalance-mitigation strategies and their effect on sensitivity and AUC.
- Target model performance of $\text{AUC} > 0.90$ and sensitivity $> 85\%$ at 80% fixed specificity, subject to dataset limits.
- A deployed web demo (Gradio or Streamlit) that accepts an uploaded dermoscopic image and returns a malignancy probability with a confidence indicator.
- A written evaluation report including per-model metric tables, ROC curves, confusion matrices, and calibration plots.
- As a stretch goal: multi-GPU data-parallel training experiments and an ensemble of the top-performing models.

7 Team Roles and Timeline

7.1 Roles

- **Bryan Torres — Model Development & Training:** implement and train ResNet-50, EfficientNet, and DenseNet; manage hyperparameter tuning and cross-validation.
- **Roman Chenoweth — Data & Evaluation:** build the preprocessing and augmentation pipeline, implement the full metric suite, produce evaluation plots, and run the imbalance ablation experiments.
- **Cian Teague — Deployment & Lightweight Models:** implement and train MobileNetV2 and ViT; build and deploy the web demo; conduct inference-time benchmarking.

7.2 Timeline

Weeks 1–2	Finalize dataset, set up training infrastructure, implement data pipeline, train ResNet-50 and EfficientNet-B0 baselines.
Weeks 3–5	Train remaining architectures; run imbalance ablation experiments; begin cross-model comparisons.
Weeks 6–8	Complete full evaluation suite; run ensemble experiments; begin web application development.
Weeks 9–11	Deploy web demo; conduct inference benchmarking; run ablations on augmentation strategy.
Weeks 12–13	Finalize evaluation report, produce all figures and tables, and prepare the final presentation.

8 Limitations and Ethical Considerations

This project is **not** intended for clinical use. Model predictions will be incorrect in some cases and may vary systematically across skin tones, image acquisition conditions, and clinical institutions not represented in our training data. We will (i) discuss dataset bias and generalization limits in the final report, (ii) avoid releasing any patient metadata, and (iii) clearly frame the web demo as a research

prototype for educational purposes only, not as a diagnostic tool. Users of the demo will be shown an explicit disclaimer stating that the model output should not be used to make medical decisions.

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